2023 AMC 10B Problem 4 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 4. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 4

Problem:

Jackson's paintbrush makes a narrow strip with a width of 6.5 millimeters. Jackson has enough paint to make a strip 25 meters long. How many square centimeters of paper could Jackson cover with paint?

Answer Choices:

A. 162500

B. 162.5

C. 1625

D. $1,625,000$

E. 16250

Solution:

Convert both measurements to centimeters. Because 1 millimeter is $\frac{1}{10}$ of a centimeter, the width of the line is $0.65 = 6.5 \times 10^{-1}$ cm. Because a meter is 100 centimeters, the length of the line is $2,500 = 2.5 \times 10^{3}$ cm. The total area is therefore

$$\left(6.5 \times 10^{-1}\right)\left(2.5 \times 10^{3}\right) = 16.25 \times 10^{2} = \text{(C)}\boxed{1,625} \text{ square centimeters.}$$

Note: This is about 2.7 sheets of letter-sized paper.

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](https://maa.org/math-competitions)

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